

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-504-1
AUTOMATIC SINGLE PHASE TESTING EQUIPMENTS	DATE: 13-12-2023

EDMS 20-504-1

TECHNICAL SPECIFICATION

FOR

AUTOMATIC

SINGLE PHASE TESTING EQUIPMENTS

ISSUE: DEC -2023-REV -1

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Specification of Automatic Single - Phase Meter Testing Equipment

1. SCOPE

This specification applies to stationary meter test units (MTUs) permanently installed in EDC laboratories' used for testing and calibration of electricity meters, in particular for their type test, acceptance test and verification test. It covers the requirements for automatic MTUs for indoor laboratory application and applies to newly manufactured MTUs to test electricity meters on 50 Hz networks with an AC voltage up to 600 V (phase to neutral).

2. STANDARDS

Table 1

Identifier	Description
IEC 62057-1:2023	Electrical energy meters – Test equipment, techniques and procedures – Part 1: Stationary meter test units (MTUs).
IEC 62052-11:2003	Electricity metering equipment (A.C.) – General requirements, tests and test conditions – Part 11: Metering equipment
IEC 60417	Graphical symbols for use on equipment (available at http://www.graphicalsymbols.info/equipment)
IEC 60617	Graphical symbols for diagrams (available at http://std.iec.ch/iec60617)
IEC 60664-1:2020,	IEC 60664-1:2020, Insulation coordination for equipment within low-voltage systems – Part 1: Principles, requirements and tests
IEC 60721-3-2:2018,	Classification of environmental conditions – Part 3-2: Classification of groups of environmental parameters and their severities – Transportation and handling
IEC 60721-3-3:2019	IEC 60721-3-3:2019, Classification of environmental conditions – Part 3-3: Classification of groups of environmental parameters and their severities – Stationary use at weather protected locations
IEC 61010-031:2015	Safety requirements for electrical equipment for measurement, control and laboratory use – Part 031: Safety requirements for hand-held probe assemblies for electrical measurement and test IEC 61010-031:2015/AMD1:2018
IEC 61010-1:2010	Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 1: General requirements IEC 61010-1:2010/AMD1:2016
IEC 61140:2016	Protection against electric shock – Common aspects for installation and equipment
IEC 61326-1:2020	Electrical equipment for measurement, control and laboratory use – EMC requirements – Part 1: General requirements
BS EN50470-1	Electricity metering equipment (A.C.) – General requirements, tests and test conditions – Part 1: Metering equipment

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BSEN50470-3	Electricity metering equipment(A.C.)-Part3: Particular requirements-Static meters for active energy (class indexes A,B and C)
IEC 62055	Part 21: Framework for standardization Part 31: Particular requirements- Static payment meters for active energy (classes 1).
BS EN 61268	Alternating Current Static VAR-Hour Meters for Reactive Energy (Classes 2 and 3)
IEC 62053-22:2003	Electricity metering equipment (A.C.) – Particular requirements –Part 22: Static meters for active energy (classes 0,2 S and 0,5 S)
IEC 62053-23	Electricity metering equipment (A.C) static watt hour meter for reactive energy meters class 2 & 3

3. ENVIRONMENTAL CONDITIONS

Ambient temperature	Maximum ambient temperature: 45 °C-(50 °C as option according EDC requirements) Minimum ambient temperature: -5 °C Maximum daily average temperature: 35 °C Maximum annual average temperature: 30 °C
Maximum relative humidity	95 %
Altitude	Up to 1000 m above sea-level

4. General:

4.1 Constructional requirements of the MTU

- The MTU shall be of protective class I according to IEC 61140:2016. All parts which are subject to corrosion under normal working condition shall be protected effectively. Any protective coating shall not be liable to damage by ordinary handling nor to damage due to exposure to air under normal working conditions. The clearance and creepage distances shall be according to IEC 60664-1:2020, overvoltage category II, pollution degree 1.
- The MTU shall be equipped with a mains switch, a mains fuse, and an emergency button.
- Single phase meter testing equipment (MTE) is required for testing and calibration of single electromechanical, static meters, prepaid meters at a time, meter types of different accuracies.
- It shall be satisfied requirements of **IEC 62057-1-2023**.

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4.2 Test voltage output terminals

- Each meter test position shall have voltage output terminals according to the number of phases. These terminals may be connected in parallel to the voltage source directly, or via MSVTs. MSVTs are specified in Annex F at IEC 62057-1-2023.
- The terminals shall be equipped with sockets of 1000 V, CAT III according to IEC61010-031:2015, 6.5.2.1 to prevent a person from accidentally touching hazardous live parts.
- The wiring shall ensure that the accuracy requirements and the reference test conditions are met at each meter test position.
- Voltage cables supplied with the MTU shall be rated for the maximum test voltages

4.3 Test current terminals

- The meter mounting system shall have current input and output terminals according to the number of phases and test positions. These terminals may be connected in series to the current source directly.
- The current terminals shall be able to carry the maximum test currents continuously, without forced cooling under the operating conditions specified. A limit on the duration of exposure should be for 15 min continuous test at least.

5. GENERAL INFORMATION

For the MTU equipment, the following general information shall be provided:

- i. Manufacturer's name or trade mark.
- ii. Designation of type, preferably in local language.
- iii. Approval mark or space for it, if requested by notified bodies.
- iv. Place of manufacture, if required.
- v. Serial number.
- vi. Protective class.
- vii. Environmental conditions for storage and operation.
- viii. Reference temperature if different from 23 °C.
- ix. Reference to instructions for installation, operation and maintenance.

5.1 The test system shall be operating under circumstance:

- i. Mains supply voltage (U_n): $3 \times 220/400$.
- ii. Variation in supply voltage is from $0.85 U_n$ to $1.1 U_n$.
- iii. Frequency f_n : 50HZ .

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- iv. Frequency range f_n is $\pm 2\%$.
- v. Power consumption shall be specified by the manufacturer.
- vi. Neutral to ground voltage shall be $< 2\text{ V}$.
- vii. Voltage quality of mains supply according as specified in EN50160
- viii. The power supply unit uses the digital switch mode amplifiers.

5.2 The test voltage circuit may provide several voltage ranges to ensure suitable load conditions and accuracy. The requirements are shown in Table 2.

TABLE 2: TEST VOLTAGE CIRCUIT

Parameter	Value
Test circuit voltage range, U	30 V to 300 V (phase to neutral)
Setting resolution	0,1 V
Accuracy of the amplitude	$\pm 0,1\%$ of the test value
Stability (S) of the amplitude	$< 100 \times 10^{-6}$ (with $T = 60\text{ s}$, $N = 10$) $< 500 \times 10^{-6}$ (with $T = 5\text{ s}$, $N = 24$)
Distortion factor at linear load	$< 0,5\%$
Maximum permitted DC voltage	$< 0,2\%$ of test voltage
Minimum output power per meter test position	15 VA RMS at the high end of the voltage range and at symmetrical loads with power factor in the range between 0,1 capacitive and 0,3 inductive
Protection	The output circuit shall be protected against short circuit and overload

5.3 Test current circuit: The test current circuit may provide several current ranges to ensure suitable load conditions and accuracy.

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THE REQUIREMENTS ARE SHOWN IN TABLE 3

Parameter	Value
Test circuit current range	1 mA to 120 A
Setting resolution	0,1 mA to 10 mA (depending on range)
Accuracy of amplitude	$\pm 1,0 \%$ ($1 \text{ mA} \leq I < 10 \text{ mA}$) $\pm 0,5 \%$ ($10 \text{ mA} \leq I < 50 \text{ mA}$) $\pm 0,1 \%$ ($50 \text{ mA} \leq I \leq 120 \text{ A}$)
Stability (S) of amplitude	$< 100 \times 10^{-6}$ (with $T = 60 \text{ s}$, $N = 10$) $< 500 \times 10^{-6}$ (with $T = 5 \text{ s}$, $N = 24$) for $I \geq 50 \text{ mA}$
Distortion factor at linear load	$< 0,5 \%$ ($I \geq 100 \text{ mA}$)
Maximum permitted DC current	$< 0,1 \%$ of test current
Minimum output power per meter test position	Minimum 30 VA RMS at the high end of the current range and at resistive loads
Protection	Output circuit shall be protected against open circuit and overload

5.4 PHASE ANGLE: the requirements for the phase angle are shown in table 4 .

TABLE 4

Parameter	Value
Range of phase angle	0° to 360°
Setting resolution	$0,01^\circ$
Stability (Sabs) of the angle setting	$0,05^\circ$ (with $T = 60 \text{ s}$, $N = 10$)
Accuracy of setting	$\pm 0,05^\circ$

5.5 FREQUENCY: the requirements for the frequency generation are shown in table 5.

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Table 5

Parameter	Value
Range of frequency	45 Hz to 65 Hz
Setting resolution	0,01 Hz
Stability (S) of the frequency setting	0,02 % (with T = 60 s, N = 10)
Accuracy of setting	±0,02 %

5.6 HARMONICS

Harmonics The superimposition of harmonics shall be possible in the voltage and the current circuits. The requirements are shown in **Table 6**

Table 6 SETTING OF HARMONICS

Parameter	Value
Harmonic range	2 nd to 21 st
Amplitude of harmonics in percentage with respect to the fundamental frequency amplitude.	0 % to 40 % for 2 nd to 5 th harmonic 0 % to 15 % for 6 th to 7 th harmonic 0 % to 10 % for 8 th to 21 st harmonic The peak value of the superimposed current or voltage may not exceed 1,4 I _{max} and 1,4 U _{max}
Setting resolution	Maximum 1 % of fundamental frequency amplitude
Phase angle with respect to fundamental frequency	0° to ±180°
Accuracy of setting	Amplitude: 1 % Phase angle: 1° related to fundamental frequency

NOTE: U_{max} and I_{max} are the specified limits of voltage and current ranges of the MTU.

6. Also it shall satisfy the following specification: -

6.1 It shall be able to supply known amount of energy to at least 20 meters the meters shall be tested using energy comparison method

6.2 reference meter and error calculation system: -

The required accuracy of the complete MTU shall be as specified in Table 19 at IEC 62057-1 (Limits of maximum permissible error (δW_{max}) of the complete MTU related to DUT).

Accuracy of Reference meter shall be (0.05) % or better for power/energy range , and 0.05% for voltage/current ranges(The values given in Table 15 IEC 62057-1-2023 apply to

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standard meters with voltage range 30 V to 300 V, current range 1 mA to 120 A and frequency range 45 Hz to 65 Hz.), directly connected in the test circuit, without using measuring transformers or transforms are incorporated in it, so that the accuracy of the reference meter is the accuracy of the whole meter test equipment .

- Reference standard voltage range :30 – 300 V (Phase – neutral).
- Reference standard current range: 1 mA – 120 A (direct).
- Current tunnel models of standard meters are not accepted.
- The reference standard meter shall have a LCD display (safely connected to the reference meter) and have a mean of display reference meter, real -time reading for voltage, current, P.F and power (P,Q,S).
- Portable reference meters are not accepted in stationary test system.

6.3 Meter suspension rack (s) :-

For at least (20) meters to be tested together, all meters are suspended at front side or at the two sides back to back. The rack(s) shall include 20 meters error display devices, and supports 20 sensors (scanning heads). Height of meter suspension bars shall be adjustable.

6.4 Sensors shall satisfy the following: -

1. Suitable for testing electromechanical, static & electronic meters. The wave length of the received signal from under test meter that can be received and evaluated by the sensors shall be within the range 500– 950 nm.
Catalogues for sensors is preferable submitted with the offer to prove compliance with the required specifications and to specify test impulse frequency for meters that can be tested by sensors.
2. Ease of alignment of scanning heads.
3. Suspension system easily adjusted for meter access and scanning head positioning.
4. Influence when the external light conditions are changing is not considerable.
5. It shall operate well in case of badly affected meter rotor marks.
6. Insensitive to sun light.
7. Scanning head devices shall be able to sense the rotor mark and led blinking of new electronic meters.
8. The Scanning head holding arrangement shall be easily adjusted vertically and horizontally.
9. The Scanning heads shall be suitable for testing both induction and static meters.

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10. The wave length of the received signal from under test static meters that can be received and evaluated by the sensors shall be within the range 500-950 nm.

6.5 Selection and adjustment devices for testing voltage / current:

The voltage and current circuits shall be independent and adjustable so as to permit all load and no- load calibration acc. To IEC “ 62053-11 “ , IEC “ 62053-21” IEC “ 62053-22” And it must have the ability of testing meters with closed link single phase “ 20 positions “ between current and voltage circuit for static and electronics meters .

6.6 console including measuring instruments, voltage connectors and current connectors.

Any MTE with design and construction performing testing functions as required in this specification (e.g. automatic electronic testing equipment) and fulfilling other items of this specifications shall be accepted .

1-Error determination of the test system:

According to IEC 62052-11 and according to accuracy offered.

2- Error indication unit :

Each measurement position shall have individual error LCD display unit. Error display unit shall be connected in such a way that any number of meters (up to 20 meters) can be tested at a time. In case one device is not working or not in use still error should be available at PC.

Other requirements:

A. The test bench:-

- Easy access to all elements is allowed.
- It shall include the necessary constructional requirements and requisite connections designed to enable meters specified above to be tested under reference conditions specified in "IEC 62052-11".
- The wiring shall be such as to minimize effects of magnetic induction, capacitive interference, and voltage drop.

B. Using 3 phase reference meter or 3 phase ICT is not accepted in single phase test benches to insure better functionality and accurate closed loop system .

C. It shall be possible to open test current circuits where current circuits, where current may be produced by induction in these circuits by voltage circuits.

D. Adjusting devices: -

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C1 -Voltage Amplifier:

- Shall permit adjusting the necessary testing voltages with 0.05% accuracy for electronic test voltage generation.
- The voltage amplifier is electronically protected against overload as well as short circuits and has a galvanic isolation between mains and output.

C2-Current Amplifier:

- Shall permit adjusting the necessary testing current (120 A) with 0.05% accuracy for electronic test current generation.
- The current amplifier shall be electronically protected against overload as well as open circuit and has a galvanic isolation between mains.

C3- Phase angle adjustment:

- The phase angle device shall permit fine adjustment of the phase angle within $0^{\circ} \dots 360^{\circ}$.
- Resolution : 0.01° .
- Setting Accuracy: 0.01° .
- Shall permit continuous adjusting of the phase angle P.F between test voltage & test current within $\pm 120^{\circ}$ (continuously step $\leq 0.01^{\circ}$).

E. Voltage and Current Amplifiers:-

-For generating (producing) test currents and test voltage shall allow compliance of M.T.E with this specification requirement.

- (separated or combined) voltage and current amplifiers are accepted if the max output current & voltage at all positions simultaneously with no influence between the accuracy of current in whole range of voltage & no influence between the accuracy of voltage in whole range of current & should be tested for the whole positions during factory acceptance test for 120 minutes
- The amplifiers shall be switch mode amplifier and transformer less.

F. Measuring parameters :

F1-The Voltage:-

Measurement of the rated voltage are within the tolerance given in the reference conditions for the specified meters within 0.05% accuracy.

F2-The Current:-

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Measurement of the rated current are within the tolerance given in the reference conditions for the specified meters within 0.05 % accuracy.

F3-The Power / Energy:-

Measuring of the power between the permissible limits for the specified meters 0.05 % accuracy.

F4- Frequency:-

The accuracy of the frequency will be within 0.01 HZ.

F5- Phase angle displacement indication or display:-

Scale graduations (if analog indication) or resolution (if digital display) shall be 0.01° , with setting accuracy 0.01° .

F6- Transmission of Meter Data:-

The test system should be provided by optional hand held terminal to collect meter related data in meter test bench and transfer it to the control software for further processing.

Test methods of calibration shall be at least:-

A) Semi-automatic operation:

1. Error calculation of the batch of at least 20 meters tested simultaneously, and error display at each meter test position by any meters error calculation system using P.C., reference standard, 20 scanning heads,.....etc.
2. Display of the specimen's error 19.99% with floating decimal point. Values superior to 19.99% will be indicated with one decimal digit, values superior to 199.9% without decimal digit.
3. i.e. error calculation by comparing the energy measured by reference meter with that of tested meters, and error calculated directly by PC or.....
4. The operator is adjusting the test console (or the testing parameters I, V, Q, may be adjusted automatically by the PC) where as the actual test procedure (error calculation, energy control, etc....) is automatically controlled.
5. Starting and no-load testing shall be allowed.
6. Testing of meters with different meter constants within the same batch shall be allowed.
7. Reference meter accuracy shall be better than or equal (0.05 %) for power / Energy

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range, and (0.05 %) for voltage /current ranges .

B) Long duration (meter register) test:

Verification of meter register (constant) using hand held terminal or Keyboard of the computer (optional hand held terminal shall be priced separately).

C) Automatic method

Automatic testing involved three activities which are one time task for a type of meter. For next time operator can use the predefined table to perform automatic testing. These modules can be run stand alone or called from any other application.

a. Type definition: - In this module operator create/edit and manage the type table as per meter type. This includes the information like nominal voltage, nominal current, maximum current, meter constant and other details. This type table will be used in automatic meter testing

b. Editor:-Here Operator defines the test sequence which are made from small steps or group of steps to perform the specific task(s)

c. Control (Automatic test):- Here operator select the desired test sequence and run the automatic testing. After initial check program will inform the maximum value of test voltage and current, which will be applied during the test. Operator is allowed to proceed or cancel the testing at any stage of testing. At the end of testing result can be stored, printout or display at the monitor.

5. Function:

MTE shall provide a complete IEC accuracy test of most known types of (single phase induction and static meters (as specified at scanning heads specification) meters. Also it shall Provide no-load, start up, and register test according to relevant IEC requirements.

6. Test circuits: -

(A) Test current circuit:-

- Current range: 1mA to 120A.
- Accuracy of current circuit: 0.05%.
- Superposition of harmonics: 2.....21th harmonics
- It is allowed to use the keys of computer keyboard instead of push buttons.

(B) Test voltage circuit :-

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- Adjustment of test voltage range (s) (nominal voltage (s)) shall allow a test voltage of 220V, + 20% , - 20 % .
- Voltage range: 30.... 300V (phase to neutral).
- Accuracy of voltage circuit: 0.05%.
- Superposition of harmonics: 2.....21th harmonics.

(C) Output Power:

- Output power for current circuit shall be 1000 VA/phase or better.
- Output power for voltage circuit shall be 500VA/phase or better.

(D) Connection of meters to be tested :-

- Manual connection shall be possible for testing more than one type of meters and the cables must be suitable to bear the flow of high current up to 120 Ampere without over heating or damage.

(E) Safety elements :-

- Necessary protection against overload and short circuiting on MTE components (e.g. reference meter, etc) shall be taken care of.

(F) A provision for testing closed link single phase meter:

A multi-secondary voltage transformers with 1 primary winding (110V...300V) and 21 secondary winding (110V...300V) which is capable of testing 20 close link meters at the same time and supplying isolated voltage for reference meter to measure the test voltage in the circuit for accuracy measurement, so single position voltage transformer is not accepted.

• Primary voltage	110V...300V, 50...60 Hz
• Secondary voltage	110V...300V
• Ratio error between primary & secondary windings	0.05%
• Ratio error between secondary windings	0.05 %
• MSVT VA	15 VA

7. Documents :-

The following documents shall be supplied with the offer:-

- A complete instructor guide or a short brief description of the test bench, describing also the modes of the whole equipment. And covering all which ensures that the offered equipment verifies specification requirements.
- Calibration (test) certificate with **ILAC** mark for reference meter from the Manufacturer lab if it is accredited according to ISO/ IEC 17025:2018 OR from an

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international lab which is accredited according to ISO/IEC 17025:2018 . The Manufacturer lab/ international lab shall submit a copy from the accreditation certificate of his lab.

- A detailed list of apparatus and accessories.
- Catalogues covering as much components of the MTE as possible to show compliance with this specification.
- This specification, each page signed by tenderer, this means that this specify is part of this offer.
- A reference list.
- The attached guarantees schedule duty complete and signed.
- Offers not accompanied with catalogues and schedule of guarantee as described above shall not be considered.

8. Testing and inspection :-

- All tests required to prove compliance with this specification requirement, shall be carried out at the expense of the supplier by the manufacturer.
- EDC has the right to send its representative(s) to attend such tests. testing dates shall be advised by the contactor to permit witnessing by EDC representative(s).
- Testing certificates shall be submitted to EDC.
- Unless otherwise accepted in writing by EDC no goods shall be shipped before inspection and testing and acceptance of EDC representative(s) for such equipment.
- Testing in the presence of EDC representative(s) shall not relieve supplier from his full responsibility about supplying goods according to all above specifications.
- Testing shall also include any test EDC shall see necessary to prove correctness of guarantee table values for the MTE or any of its components, especially the VA power output of the testing current and voltage circuits (e.g. testing of voltage transformer and phase shifter by loading them by the guarantee table values of VA for a long period regardless the number of test meters specified as 20 meters) also electric generators VA may be tested.

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TABLE "1"

Currents that the meter testing equipment shall be able to test

According to IEC 62052

Following current values shall be indicated digitally or on the graduated scale.

Meter Ib/Im	10/30	10/40	40/80
Testing current .005 Ib	.05	.05	0.2
.05 Ib	0.5	0.5	2.0
0.1 Ib	1.0	1.0	4
0.2 Ib	2.0	2.0	8
0.5 Ib	5.0	5.0	20
1.0 Ib	10	10	40
0.5 Im	15	20	40
1.5 Im	30	40	80

Ib: basic current of the meter.

Im: maximum current of the meter.

8- GUARANTEE:

- Supplier (contractor) shall guarantee supplies against all defects arising out of faulty design manufacture, or defective material for three years from date of delivery to EDC stores for supplier's with previous supply to EDC .
- In case of supplying for the first time, the guarantee period shall be extended to five years at least from delivery date.

9. EQUIPMENT AND ACCESSORIES:

- a) The various sub-assemblies shall be easy access to facilitate maintenance.

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- b) The metallic equipment shall be treated with anti-corrosion.
- c) The testing equipment shall constitute all equipment and accessories needed to perform tests as mentioned above including (the test voltage cables , test current cables , set with pluggable pins for testing currents to 120 & 60 A .
- d) The following components of the MTE shall be manufactured by MTE manufactured.
 - 1- Reference standard meter.
 - 2- Scanning heads and error indicators.
 - 3- Current & voltage amplifiers (the amplifiers shall be transformers less)
 - 4- Frequency waveforms generation.
- e) The test bench has the capability to test other portable reference meters with different accuracies.

N.B.:

- The test bench and the amplifiers manufacturer should be the same manufacturer of the reference standard.
 - The manufacturer should have a reference list for the delivery of the same quoted bench to any Electrical utility in Egypt and he should supply us with a copy of the purchasing order together with a field visit for the unit mentioned in the purchasing order.
 - The Manufacturer should have a reference list for the delivery of a stationary test bench to any electrical utility in Egypt.
 - The Bidder shall submit the three (3) satisfactorily performance report for the test equipment having satisfactory operation of the smiler tender equipment or better equipment from any international lab.
 - The bidder shall submit the three (3) Satisfactorily performance report for the test equipment different accuracies from any electrical utility in Egypt.
 - The supplier must provide proof of technical competence to repair MTU after the end of the warranty period, such as providing maintenance orders and maintenance contracts for this type of MTU for worldwide with the possibility of field visits to sites under the maintenance of the supplying company.
- 2- The country of origin should be Europe, Japan, or USA and the manufacturer should submit a certificate of origin and arrange a training visit and an inspection visit for at least two engineers.

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- 3- The manufacturer should have an agent in Egypt with original S14 certificate; the agent should submit the original S14 certificate on request for matching with submitted copy.

TABLE "X"

Reference conditions

Influence quantity	Reference value	Permissible tolerances for meters of class		
		0.5	1	2
Ambient temperature	Reference temperature or in its absence 23° C (1)	± 1° C	± 2° C	± 2° C
Working position	Vertical working position (2)	± 0.5°	± 0.5°	± 0.5°
Current / Voltage	Reference Current / Voltage	± 0.05%	± 1.0%	± 1.0%
Frequency	Reference Frequency	± 0.2%	± 0.3%	± 0.5%
Wave form	Sinusoidal voltage and current	Distortion factor		
		2%	2%	2%
Magnetic induction of external origin at the reference frequency	Magnetic induction equal to Zero	Induction value which causes a variation of error not greater than (3)		
		0.1%	0.2%	0.3%

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(1) If the tests are made at a temperature other than the reference temperature, including permissible tolerances, the results shall be corrected by applying the appropriate temperature coefficient of the meter.

(2) Determination of the vertical working position (see sub-clause 5.2).

The construction and assembly of the meter should be such that the correct vertical position is ensuring (in both the front-to-back and left-to-right vertical planes) when:

- The base of the meter is supported against a vertical wall
- A reference edge (such as the lower edge of the terminal block) or a reference line marked on the meter cause is horizontal.

10. TECHNICAL SPECIFICATION

No. Of tested meter	20
Supply	3×230/400 V + 10%, -15% , 50Hz ± 2 %
Voltage frequency	50Hz ± 2 Hz
Test voltage Circuit:	30...300 (P-n)
- Voltage range	500VA / phase
- Power	Against short circuits & overload
- Protection	≤ 0.05%
- Accuracy	
Test current circuit :	1 mA...120A
- current range	1000 VA / phase
- Power	Against open circuits
- Protection	≤ 0.05%
- Accuracy	
Reference meter with display:	Not more than 0.05
- Accuracy of power	
- Measurement current range	%
- Accuracy of current range	
- Measurement voltage range	1 mA...120A

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- Accuracy of voltage range	$\leq 0.05\%$
	30...300V
	$\leq 0.05\%$
Phase angle: - Phase shifting - Resolution and accuracy - Step	From 0° to 360° 0.01° 0.01°
Harmonic generation of current and voltage amplifiers: - Harmonics	2 nd to 21 th
Connection cables: - Type of connection - Connection with source - Connection with PC	Free wires Yes Yes
Operating conditions: - Temperature - Relative humidity	From 0° to 50° Up to 95%

11. Test bench software capabilities

The software of the test bench should have the next capabilities:

- Pre-warming
- Accuracy testing of energy meters
- Starting Current Test
- Creep Test

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- Influence quantity test as per IEC 62053-11,21, 22,23 (which are mainly linked with accuracy test)
- Dial test
- Register test
- Influence of self-heating
- Influence of heating
- Harmonics generations (Harmonics up to 2nd to 21th).
- Voltage interruption/dip test according IEC 62052-11
- Absolute measurement or Testing of reference meter class 0.05

12. MARKING

- Each test equipment shall be provided with an easily readable nameplate
- Each nameplate will include at a minimum the following:
 - i. No. of positions
 - ii. manufacturer's name
 - iii. Country of Origin
 - iv. date of manufacturing
 - v. accuracy class.
 - vi. serial number.
 - vii. month and year of manufacturing (not coded).
 - viii. number of phases.
 - ix. rated VA.
 - x. rated current.
 - xi. rated voltage.
 - xii. rated frequency

13. Tests

- All tests shall be according to IEC62057-1-2023

14. Training

15. TRAINING and CUSTOMER SUPPORT:

- The supplier shall provide adequate on-site training for a sufficient number of...EDC representatives, to be agreed by (...EDC) and supplier.
- The training should include and not limited to the following:
 - i. Installation and Programming.
 - ii. Testing and Commissioning.

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- iii. Troubleshooting.
- iv. Operating and Maintenance
 - The Supplier must have a dedicated service center and support line in Egypt to cover Test equipment's benches.
 - EDC reserves the right to claim a free after-sales service for the entire warranty

Period. After-sales service should include and not limited to the following:

- i. Technical support.
- ii. Maintenance.
- iii. Original Spare parts.
 - All services, spare parts, materials and accessories required during the after-sales

Maintenance should be delivered by the supplier immediately wherever required

(Warehouse, site, laboratory, etc.)

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16. Guarantee Table: MTE single phase Meter

1. Type and model.....
2. Manufacturing Company.....
3. Country of manufacture
4. Standards which testing equipment satisfies
5. Test current circuit:-
 - ❖ Current output VA
 - ❖ Current range A
6. Test voltage circuit :-
 - ❖ Voltage output (voltage circuit output power) VA
 - ❖ Voltage range V
7. Phase adjustment :-
 - ❖ Stepless adjustment of power within zero to $\pm$ Degree
 - ❖ Operation of phase angle by (hand /motor)
 - ❖ Indication of phase angle by (KW meter or phase angle meter) .
 - ❖ Quick selector of $\cos \phi = 1.0$ to $\cos \phi = 0.5$ YES () NO ()
8. Reference standard meter :-
 - ❖ Type, model Manufacturing Company.....
 - ❖ Accuracy Active energy (P.F = 1)
 - Active energy (P.F = 0.5).....
 - ❖ Error calculation system :-
 - ❖ Type, model, Manufacturing Company.....
 - ❖ Capacity (active meters).....
9. Suspension Rack :-
 - ❖ Built in or separate

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- ❖ Capacity: number of meters to hang
- ❖ Height of the horizontal bar adjustment range
- ❖ Has a rail to carry scanning heads..... YES () NO ()
- ❖ No. of scanning heads supplied with rack.....
- ❖ No. of individual indicators at test positions.....

10. Scanning heads for testing meters of mech. type :-

- ❖ Type, model, Manufacturing Company.....
-
- ❖ Ease of alignment of photo-cell pick-up , pick-up arms easily swung YES () NO ()
out for meter access.....
- ❖ Scanning heads suitable for mech. and static meters..... YES () NO ()
- ❖ Focal distance from sensor to meter..... Mm
- ❖ Heads insensitive to sun light..... YES () NO ()
- ❖ Diodes colour (Green is preferable)

11. Main Connection :-

- Un stabilized mains , 50 HZ..... Volt

12. Scanning heads for testing KWH – meters of mech . type :

- Type
- Ease of alignment of scanning heads, suspension system easily adjusted for meter access and scanning head positioning.YES ()NO()